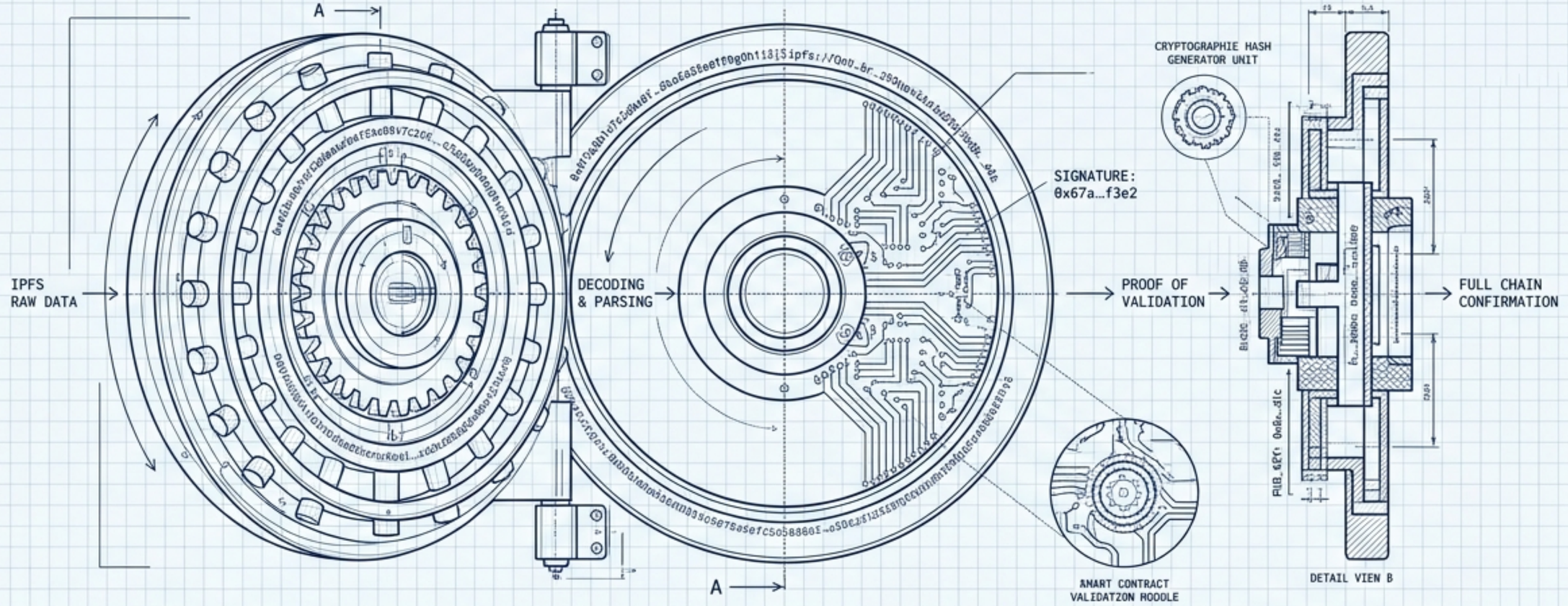
The Auditor's Field Guide to the Commons Vector Protocol

A sequential forensic pathway from raw IPFS download to full chain validation.



Version: CVP-0.1 | Architecture: Content-Addressed Object and Provenance Protocol | Trust: commons-vector-v1

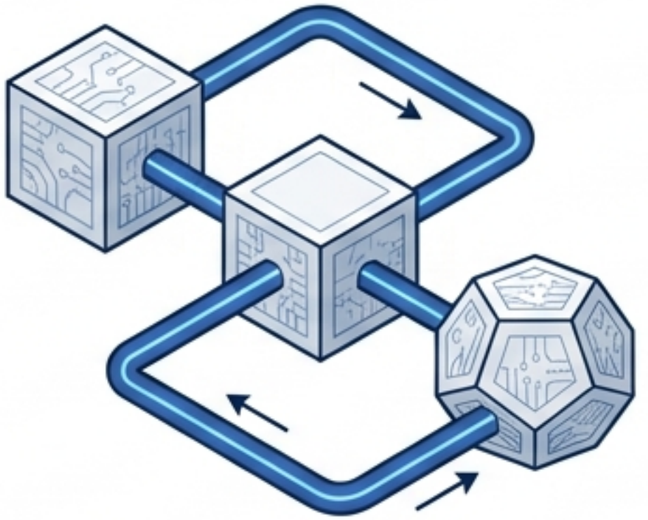
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Verification is an act of arithmetic, not faith.



Traditional verification relies on trust in an authority, physical stamps, and vulnerable paper records. It asks:

Who says this is real?



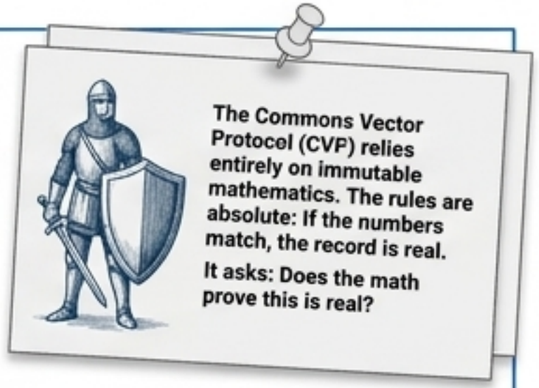
The Commons Vector Protocol (CVP) relies entirely on immutable mathematics. The rules are absolute: If the numbers match, the record is real. It asks:

Does the math prove this is real?

> **“The first record wins. Not by force, not by law, but by time. Because the record is immutable and linked, no one can go back in time and change what is already written.”**

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The Anatomy of a Content Addressable Record (C.A.R.)



Example: Visual Instrument Payload

Zone 1: Content (CID)

`bafyb...5zbfe` — The immutable IPFS hash of the payload.

Zone 2: File-ID3 Vectors

The Spatial (where) and Rule (how) vectors defining the object's context.

Zone 3: Composite Key

[hash(cid + spatial + rule)] — The unique ID of the record itself.

```
{
  "cid": "bafyb...5zbfe",
  "spatial": [12.34, 56.78],
  "rule": "access-policy-v1",
  "composite": "[hash(cid + spatial + rule)]",
  "sig": "Ed25519_signature_string",
  "seq": 1,
  "prev": "[...]"
}
```

Zone 4: Signature

Ed25519 authorization from the key holder.

Zone 5: Provenance

seq: 1, prev: [...] — The cryptographic link to the prior record in time.

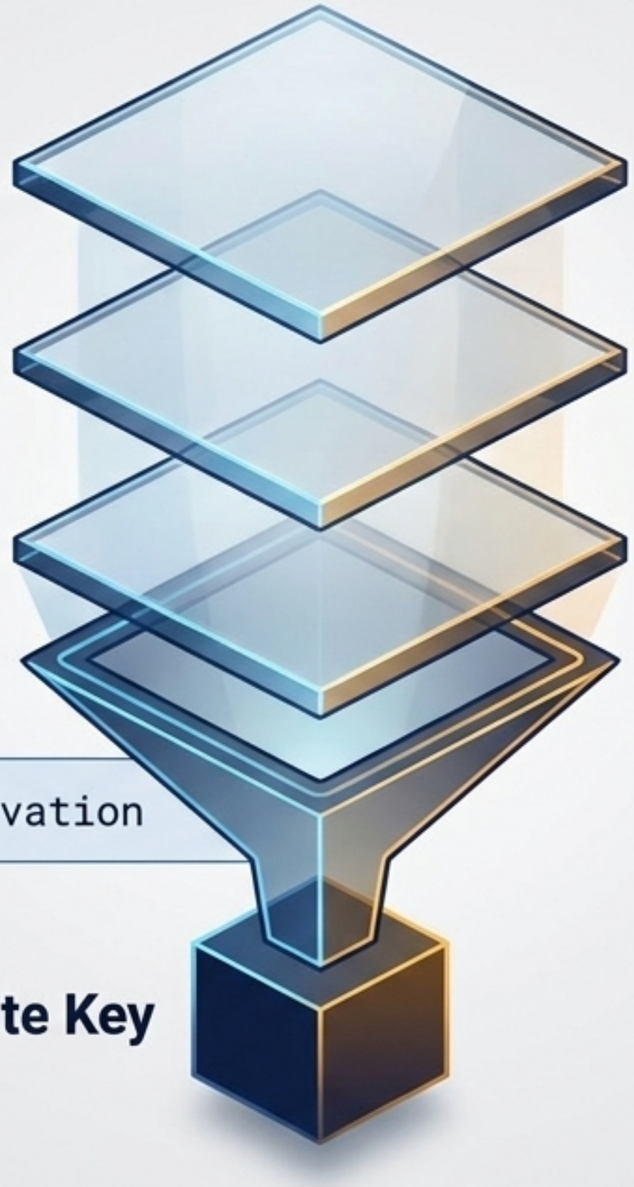
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The Core Identity: The Tri-Vector Architecture

Content Vector (CID)
Represents WHAT it is (e.g., hash of the payload).

Spatial Vector (V1)
Represents WHERE it lives in the namespace (e.g., trust:commons-vector-v1:living:witness:1).

Rule Vector (V2)
Represents HOW it is governed (e.g., dictionary: ILPJ-v1.0.4).

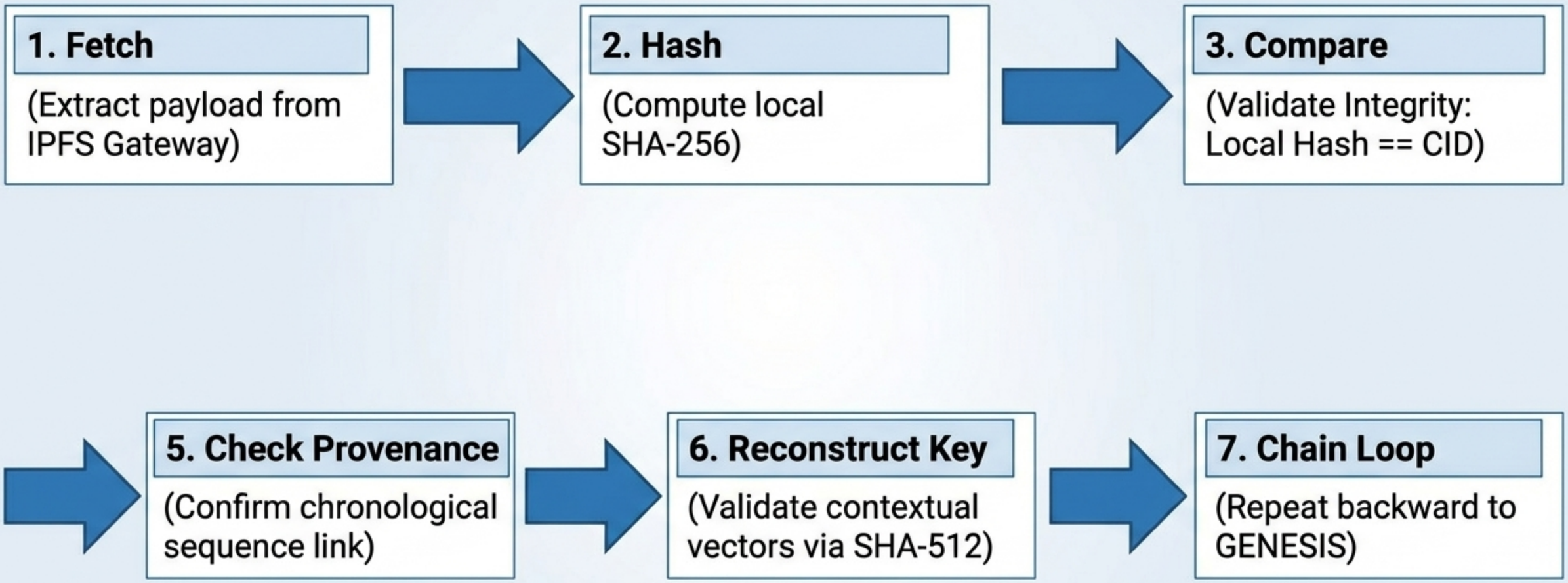


These dimensions must never be conflated. When hashed together, they generate the

When hashind. When hashed together, generate the Composite Key: the unique "number" of the record that identifies it by both its content and its declared context.

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The End-to-End Verification Pipeline



Note: The output of one step becomes the exact cryptographic input for the next.

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Step 1: Fetch the Data Payload

To begin the audit, the record must be extracted from the InterPlanetary File System (IPFS) to your local machine. You only need the Content Identifier (CID).

Gateway

https://yellow-managerial-leopon-627.mypinata.cloud/ipfs/

+

CID

[bafyb...xjmy2]

=

Execution

Open URL in browser to download the raw JSON payload.

The CID specifies WHAT the content is, not where it is stored. It can be fetched from any IPFS node holding a copy.

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Steps 2 & 3: Hash and Compare for Absolute Integrity

Step 2: Local Hash Computation

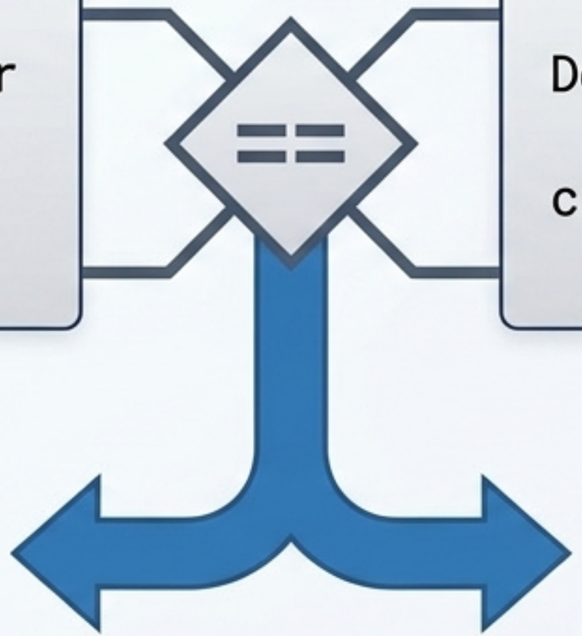
```
> sha256sum witness-ktrg-1.car

Output: bafy...5zbfe
```

Step 3: Target Declaration

```
Declared cid in JSON:

cid: bafy...5zbfe
```



IF **MATCH**:
The file is exactly as it was upon upload. Not a single byte has been altered. **Integrity confirmed.**

IF **MISMATCH**:
The file has been tampered with or corrupted. The record is invalid. Halt audit.

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Step 4: Authenticate the Cryptographic Signature

Message: The record's Composite Key

Public Key: The public_key field declared in the JSON

Signature: The value field declared in the JSON



TRUE / FALSE

Pass these three elements through a standard Ed25519 verifier. This math proves exclusively that the holder of the specific private key authorized this exact record.

You do not need to trust who they say they are. You only verify that the key holder signed the record.

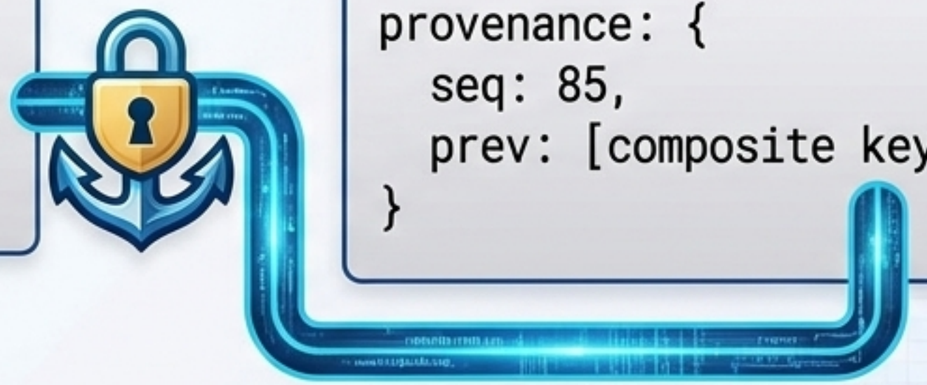
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Step 5: Verify the Provenance Link

CVP records do not exist in isolation. They form a linear, unbroken history. Every record must point to the precise mathematical identity of the record that came immediately before it.

Sequence 84

```
number: {
  composite: [composite key of seq 084]
}
```



Sequence 85

```
provenance: {
  seq: 85,
  prev: [composite key of seq 084]
}
```

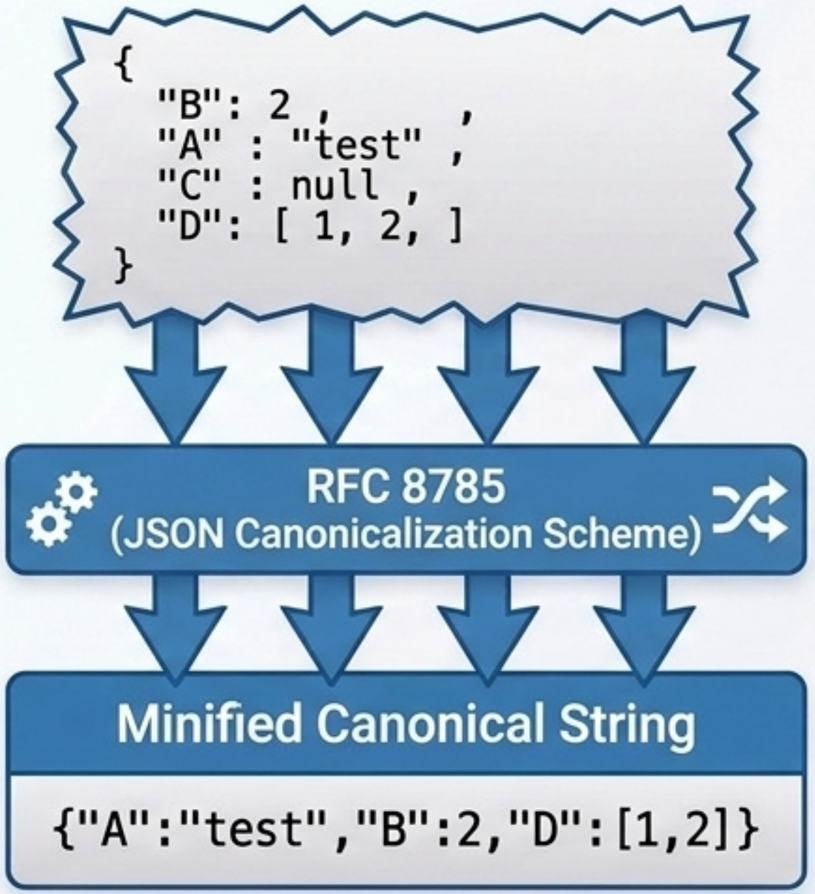
RULE: If the ‘prev’ field does not perfectly match the prior record’s Composite Key, the timeline has been manipulated.

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Essential Pre-Condition: Canonical Serialization

Before an auditor can mathematically reconstruct the Composite Key (Step 6), the data must be perfectly formatted. A single stray space would change the hash.

Unstructured JSON



The Rules of RFC 8785

- 1. Object keys MUST be sorted in lexicographic (alphabetical) order.
- 2. No insignificant whitespace or trailing commas.
- 3. Numbers MUST use the shortest possible representation.
- 4. Null values are excluded.

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Step 6: Reconstruct the Composite Key

An attacker might swap a file's payload but keep the signature. To prevent this, the auditor must manually recalculate the Composite Key to prove the Tri-Vector context is genuine.



Compare your Calculated Key to the composite value declared in the file. A match proves the record's context is mathematically bound to its payload.

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Step 7: The Chain Loop Back to Genesis

A single verified record is just a point in time. A fully verified audit requires repeating Steps 5 and 6 sequentially backward through the entire chain.



End Condition:

- Repeat process for Sequence N down through 1.
- The loop terminates successfully when you hit seq: 0 where prev: GENESIS.

If the loop reaches Genesis without a mismatch, you have mathematically proven an unbroken, tamper-evident history of time and state.

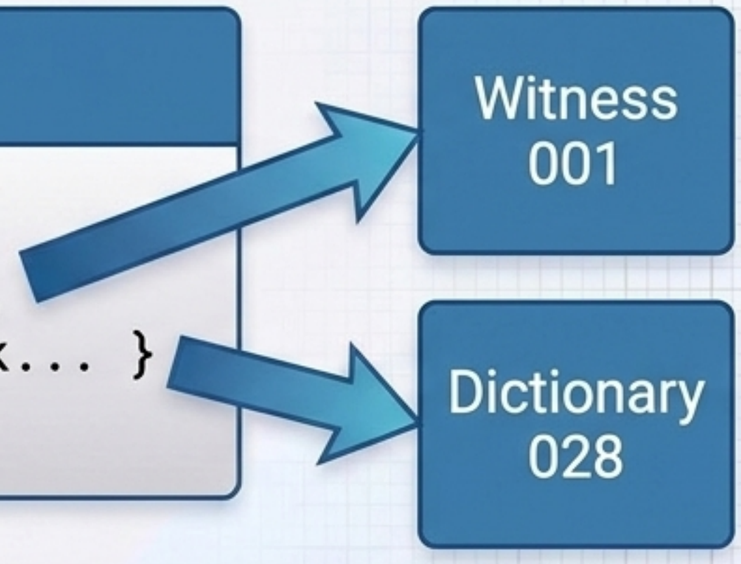
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The Insertion Protocol: Auditing Composite Documents

An instrument does not need to be copied into a document. It only needs to be named by its Composite Key. The chain is not extended by an insertion; it is only referenced.

Deed of Trust (Master Wrapper .car)

```
insertions: [
  { car: 001, label: Witness 1, composite: a4f9... },
  { car: 028, label: Dictionary Root, composite: bafk... }
]
```



The **Auditor's Rule:** To audit a nested document, the verifier must pull the CID for EVERY inserted composite key and run the 7-step pipeline on all of them. If one fails, the master document is invalid.

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Automating the Audit: The Verifier Script

While the manual steps prove the logic, auditors use scripted automation to validate entire trust chains instantly.

```
// Initialization
prev = "GENESIS"
for i in 1..N:
    record = fetch(seq_i)
    // Step 2 & 3: Integrity
    assert hash(record.payload) == record.cid
    // Step 6: Context
    assert SHA512(canonical(record)) == record.composite
    // Step 4: Authorization
    assert verify_ed25519(record.signature, record.composite)
    // Step 5: Provenance
    assert record.provenance.prev == prev
    prev = record.composite
```

If the loop completes without throwing an assertion error, the entire chain is cryptographically intact.

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Diagnostic Matrix: Protocol Assertions vs. Legal Reality

The audit verifies mathematics. It does not magically create legal standing. The exact boundaries of the proof must be recognized.

Cryptographic Reality (What the Audit Proves)	Legal Fiction (What it DOES NOT Prove)
Integrity: The content has not been altered by a single byte since upload.	The content inside the file is factually true.
Authorization: Node A (the specific key holder) signed this record.	Node A has the legal authority to bind the referenced entity.
Context & Jurisdiction: This record is governed by rule-set R.	Rule-set R is legally recognized in external Jurisdiction J.

Cryptography proves content integrity, signature authenticity, and unbroken sequence. The **math proves presence.** The first record wins.

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